

GEEK ON A LEASH

Rust at the boundary between
intent and motion

OPEN SOURCE • MIT

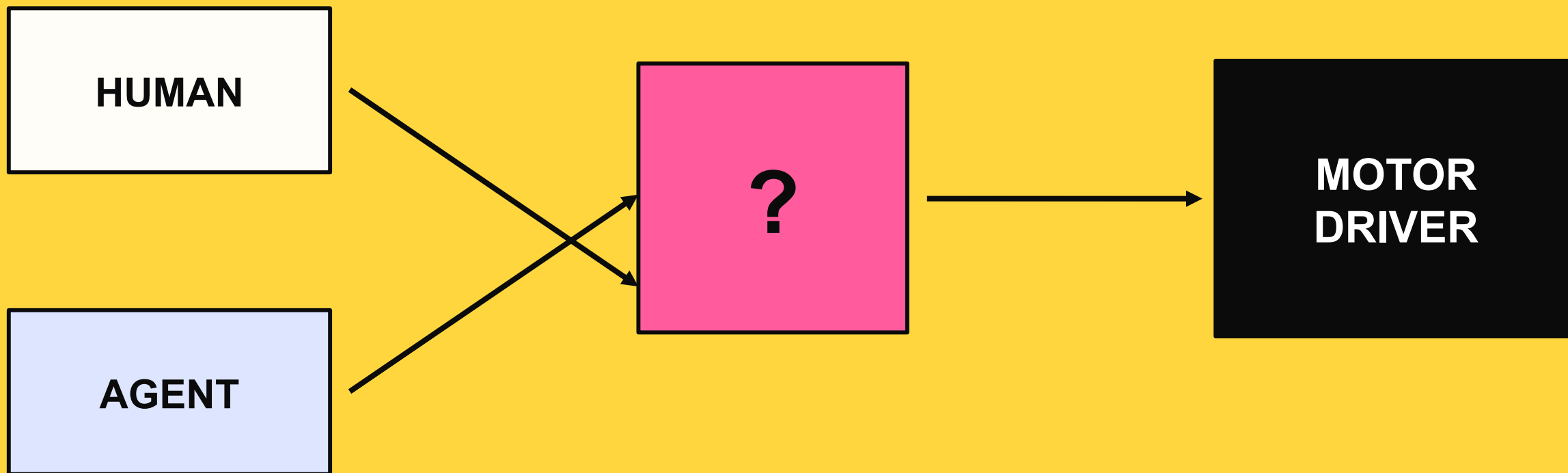


ERIC MANGANARO

Leash + Pinkie • 60 minutes



Who gets to write the motors?



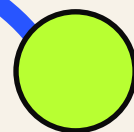
The dangerous bug is not bad planning.
It is ambiguous authority.

A leash does not choose the walk.

It limits how intent reaches the body.



intent



motion

The planner
proposes.
The boundary
decides.

SMALL • FAST • AUDITABLE

“ **AI may request motion.**

**LEASH DECIDES WHETHER
IT IS ALLOWED.**

This sentence is the architecture test.

If a feature violates it, the feature belongs somewhere else.

Pinkie is where the abstractions meet friction.

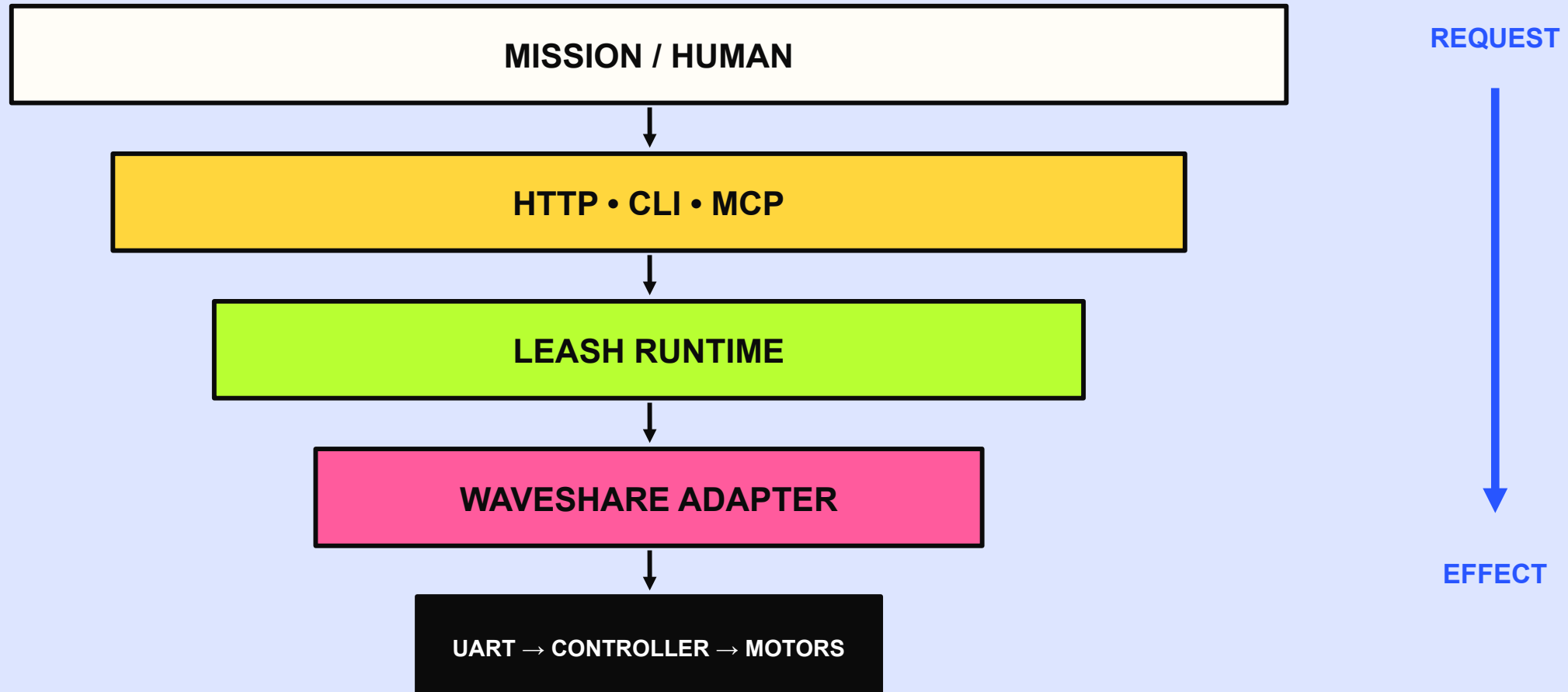


LIVE CAMERA • 2026-09-01

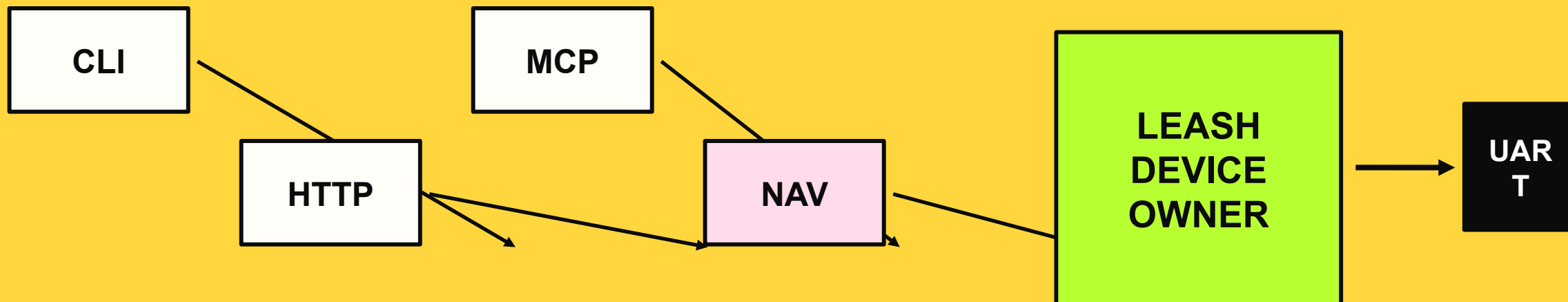


Waveshare UGV
Jetson Orin NX
LiDAR + camera + IMU + odometry

Every layer narrows what the next layer may do.



One serial owner. Zero mystery writers.



All roads converge before bytes reach the controller.

Three protocols. One capability contract.

```
$ leash drive --linear 0.08 --for 400ms
```

```
POST /v1/capabilities/drive/propose
```

```
tools.call("drive.propose", candidate)
```



```
Capability< Candidate → Authorized >
```

Syntax changes. Authority does not.

Policy above mechanism. Evidence beside both.

INTENT

mission • operator • autonomy

CAPABILITY

resolve • validate • authorize

SAFETY KERNEL

deadline • E-stop • verified zero

ADAPTER

encode • transmit • read back

HARDWARE

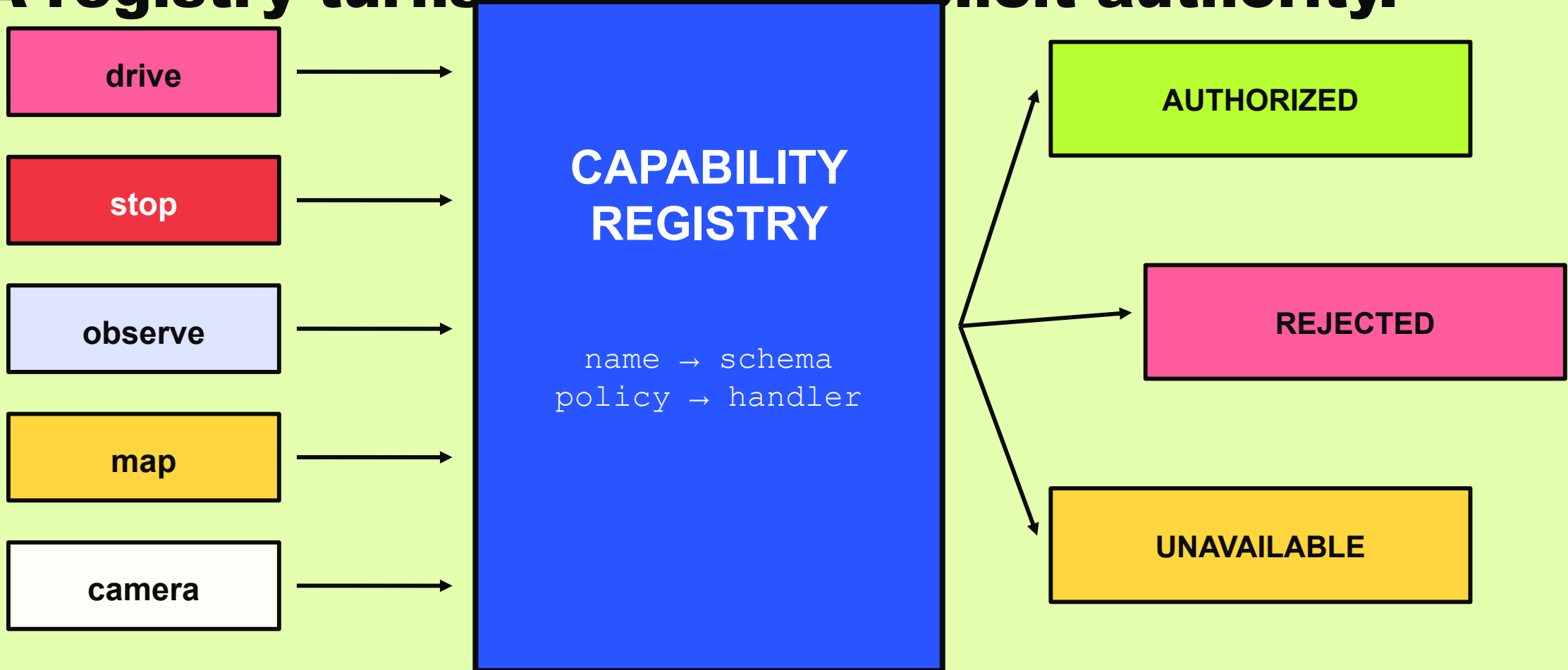
controller • motors • sensors

EVIDENCE

proposal
decision
transition
telemetry
result

Append-only, not an afterthought.

A registry turns verbs into explicit authority.



Supertraits describe the hardware boundary.

```
pub trait ControllerIo:  
    Read + Write + Send {}  
  
impl<T> ControllerIo for T  
where  
    T: Read + Write + Send  
{ }
```

Read + Write

the byte-level contract

Send

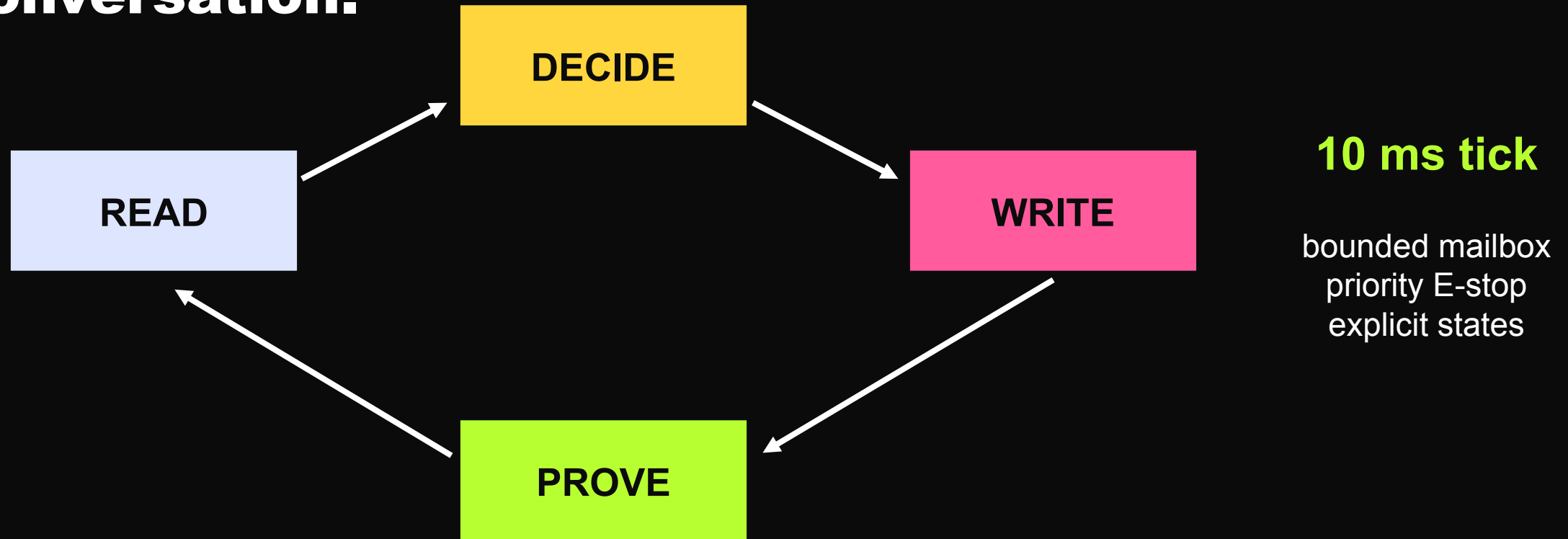
safe to move into owner thread

blanket impl

any conforming I/O becomes usable

ZERO METHODS • STRONG BOUNDS

The safety kernel is a deterministic loop—not a conversation.

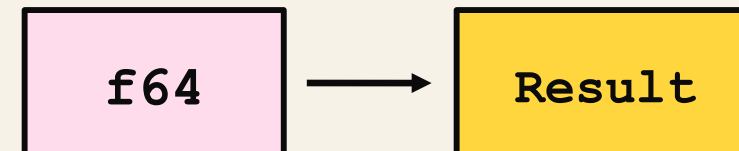


A model may miss a token deadline.
The motor boundary may not.

Smart constructors close numeric ranges.

```
pub struct NormalizedDrive(f64);

impl NormalizedDrive {
    pub fn new(value: f64)
        -> Result<Self, DomainError>
    {
        if !value.is_finite() { ... }
        if !(-1.0..=1.0).contains(&value)
    { ... }
        Ok(Self(value))
    }
}
```



private field

outside modules cannot bypass `new()`

finite + bounded

NaN and $\pm\infty$ never reach encoding

INVALID AT CONSTRUCTION

Generics encode the authority transition.

```
pub struct Candidate<C> {  
  pub deadline:  
    MonotonicNanos,  
  pub command: C,  
}
```

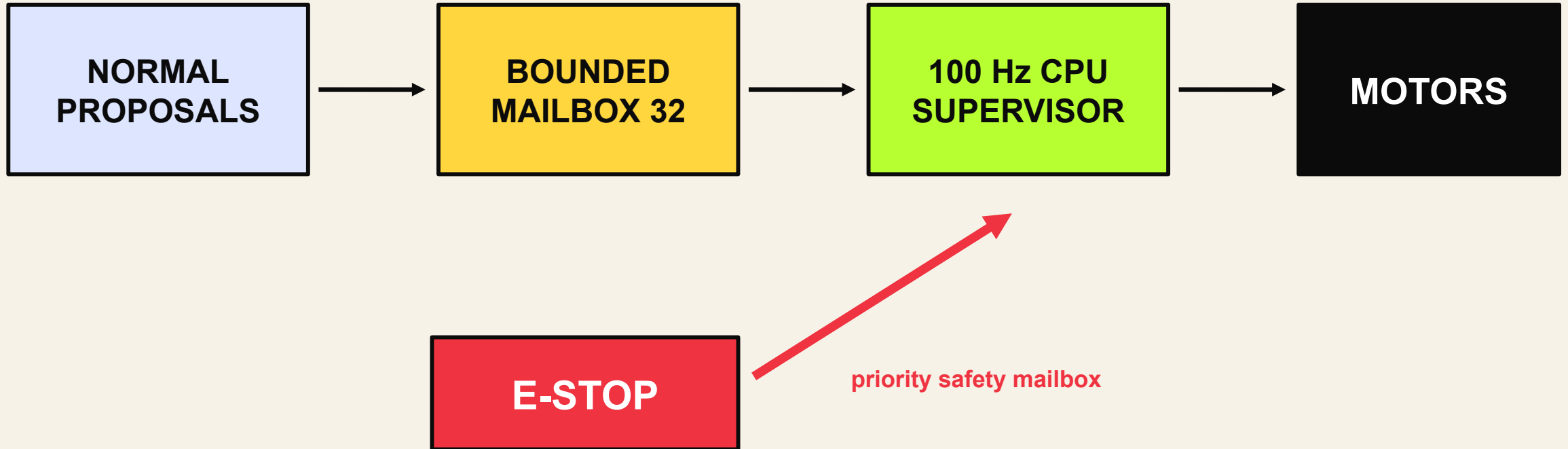
```
SafetyGate  
.authorize(candidate, now)
```

```
pub struct  
  Authorized<C> {  
    command_id:  
      CommandId,  
    evidence_id:  
      EvidenceId,  
    authorized_at:  
      MonotonicNanos,  
    command: C,  
  }
```

```
Err(SafetyDenial::Expired)
```

C is generic: the transition works for any command payload.
Authorized fields are private: gateways cannot forge the post-policy state.

E-stop skips the queue. CPU keeps authority.



Safety traffic is not ordinary traffic.

Associated types keep adapters concrete.

```
pub trait ActuationPort: Send + 'static {  
    type Acknowledgement:  
        ActuationAcknowledgement;  
    type Error:  
        Display + Send + 'static;  
  
    fn submit_drive(  
        &mut self,  
        command: Authorized<DifferentialDrive>,  
    ) -> Result<(), Self::Error>;  
}
```

&mut self

exclusive mutable access

Self::Error

adapter-specific, statically known

Authorized<...>

raw candidates cannot cross

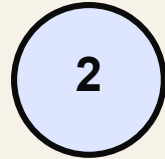
DYNAMIC HARDWARE • STATIC CONTRACT

Freshness is a first-class input to motion.



LIDAR

range + clearance



CAMERA

context + evidence



IMU

orientation + dynamics



ODOMETRY

motion + displacement



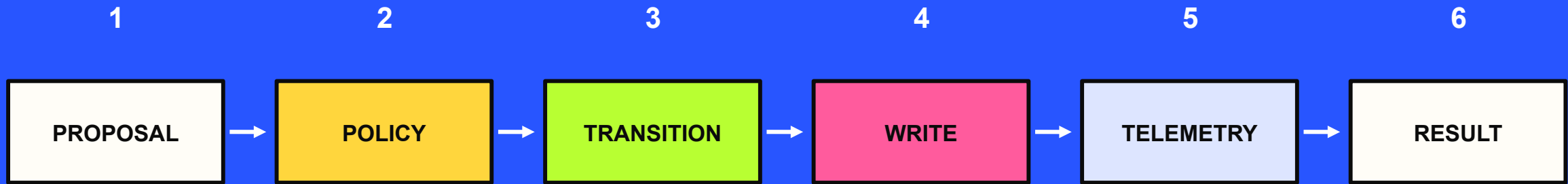
BATTERY

83.3% live

STALE ≠ SAFE

A value without an age is not a control input.

Replay should explain the crossing—not invent a story.

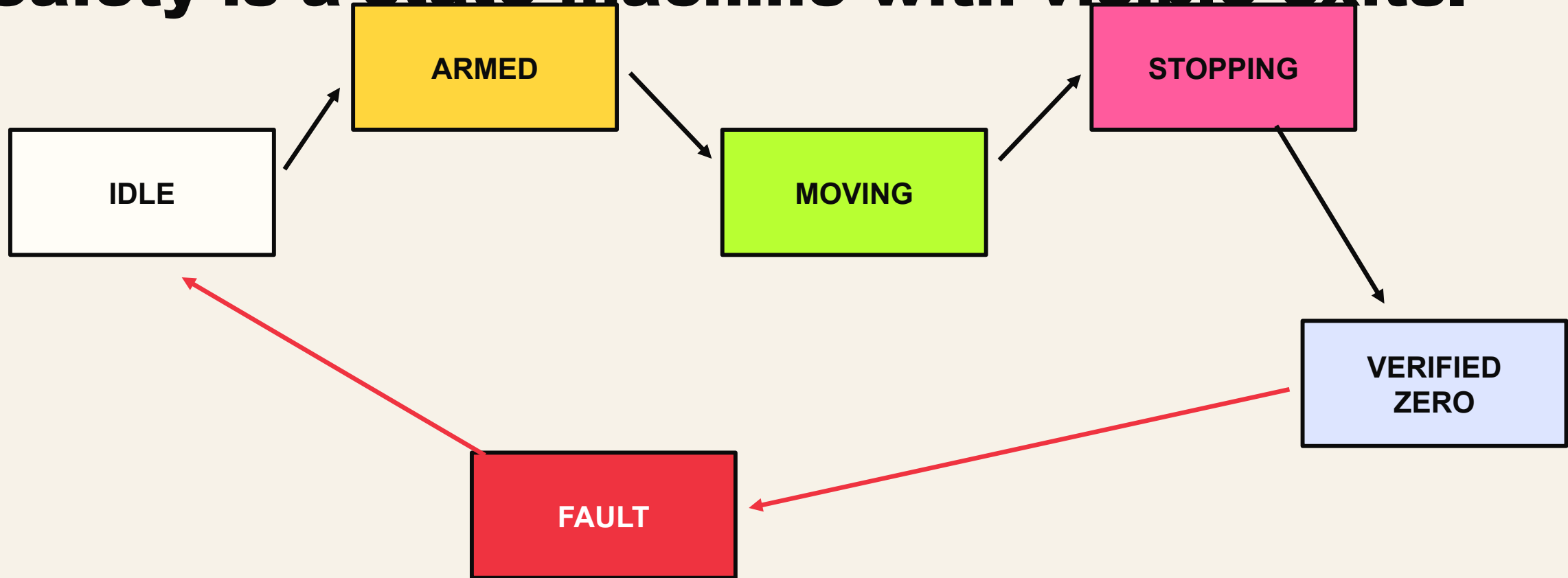


4,807,319

durable records on live Pinkie

Evidence is useful only if it survives
the process that made the decision.

Safety is a state machine with visible exits.



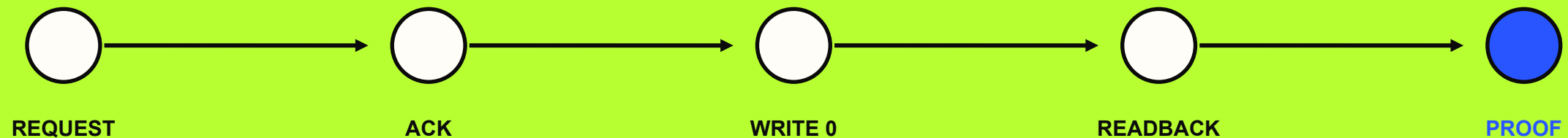
timeout • E-stop • stale sensor • ownership loss

A stop command is a request. Zero is the proof.

37.622 ms

The evidence bundle also verified final zero.

physical E-stop acknowledgement



MEASURED ON JETSON ORIN NX

CUDA accelerates. CPU owns motion.



promote only after parity + performance evidence

Phantom types stop coordinate frames from drifting.

```
pub enum Map {}  
pub enum Odom {}  
  
pub struct Frame<Tag> {  
  name: FrameName,  
  marker: PhantomData<fn() -> Tag>,  
}  
  
pub struct Pose2<Tag> {  
  frame: Frame<Tag>,  
  x: Meters, y: Meters, yaw: Radians,  
}
```

Pose2<Map>

Pose2<Odom>



won't unify

Nav2 path → typed proposal →
Leash

The boundary is small enough to measure.

58,306 ns

p99 CPU transition

0

deadline misses at 100 Hz

110,293/s

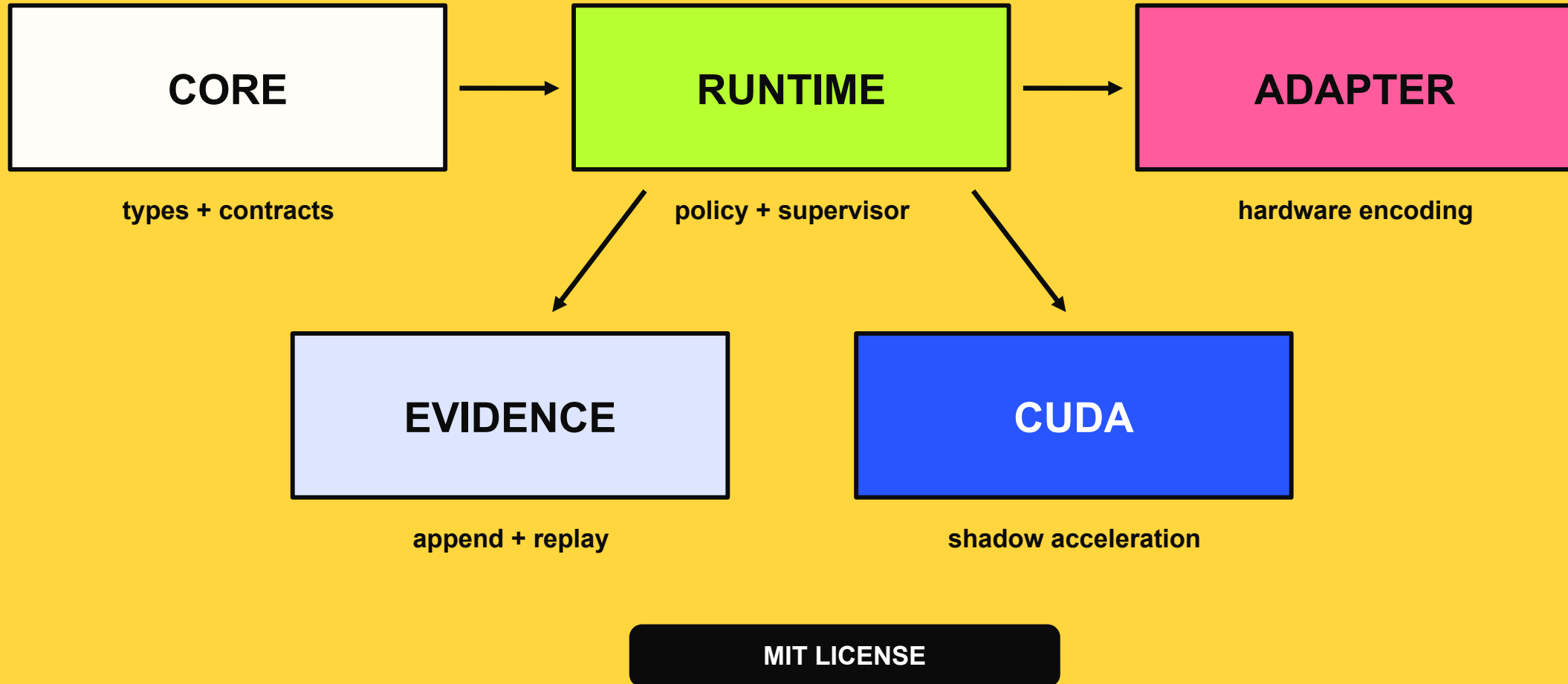
durable evidence records

37.622 ms

physical E-stop acknowledgement

JETSON ORIN NX • RECORDED EVIDENCE • NOT A SIMULATION

Leash is a boundary you can replace piece by piece.



Live today is not the same as claimed tomorrow.

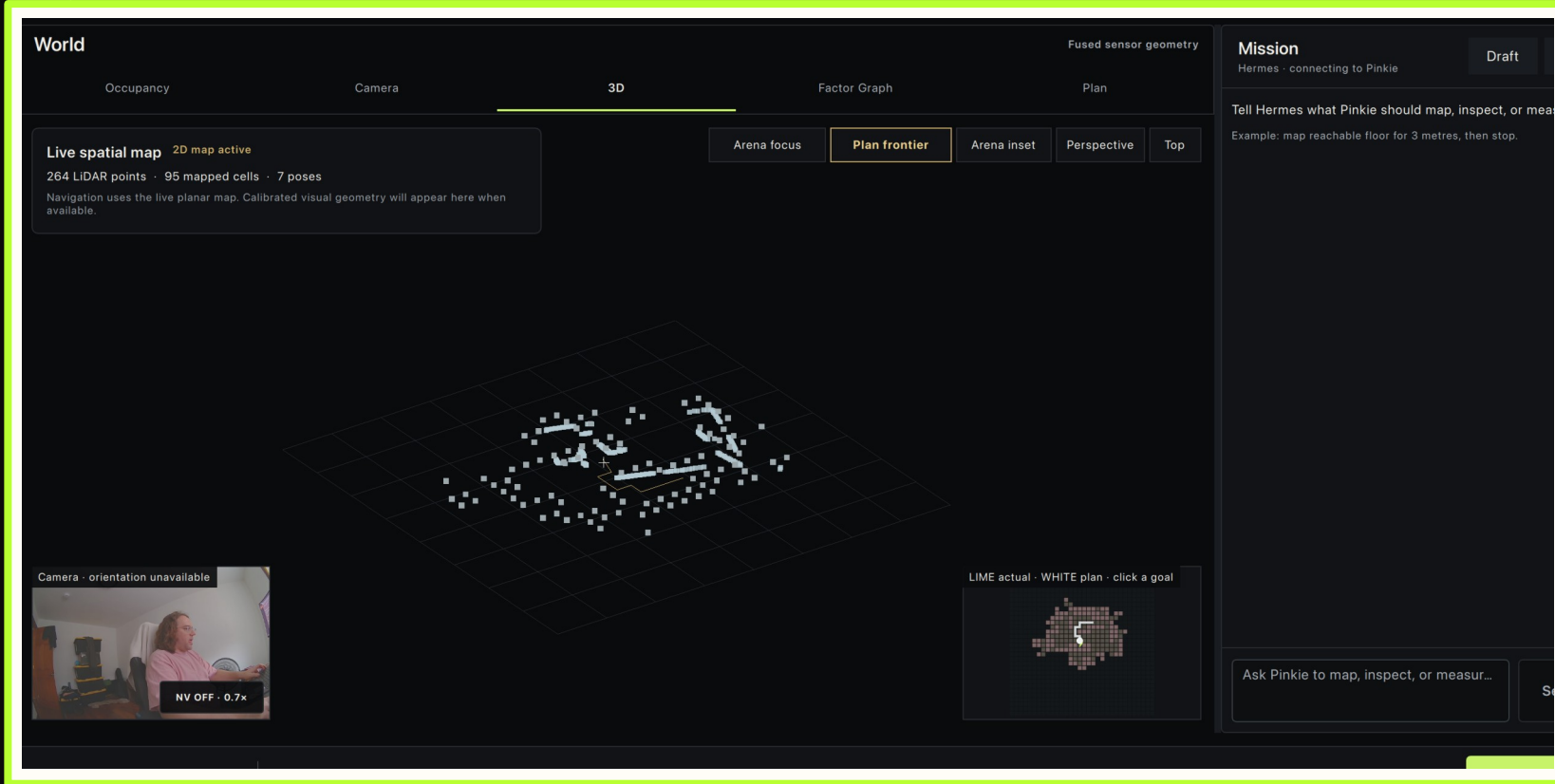
LIVE + VERIFIED

-  controller ownership
-  CPU final authority
-  fresh LiDAR / IMU / odometry
-  camera snapshot
-  CUDA available in shadow
-  durable evidence

NOT CLAIMED

-  whole-house autonomy
-  active mapping at this moment
-  visual odometry lock
-  GPU motor authority
-  RL policy in the control loop
-  semantic room understanding

Qualia thinks on a slower clock.



LEASH
10 ms



QUALIA
seconds →
minutes

scene • mission • ontology • learning

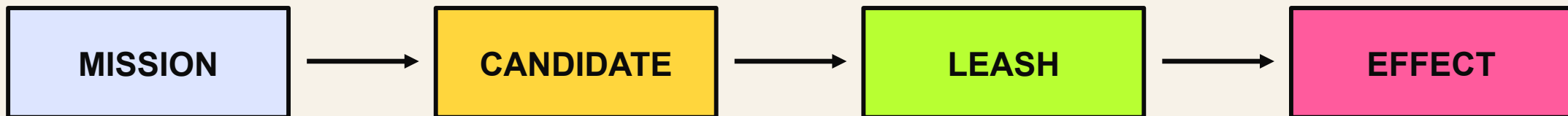
Intelligence crosses as a typed proposal.

```
1.4) - likely about to head back. Still no arena-level mission
any of it, and vision tracking remains the weak link (confidence
still not locked). Fresh3camera frame is cached at
~/hermes/image_cache/img_e39f04b54666.jpg if you want to eyeball
it's seeing.
ready | deepseek v4 flash | 57.1k/1m      - ...-planner (...mission)
—
> i want you to explore the area

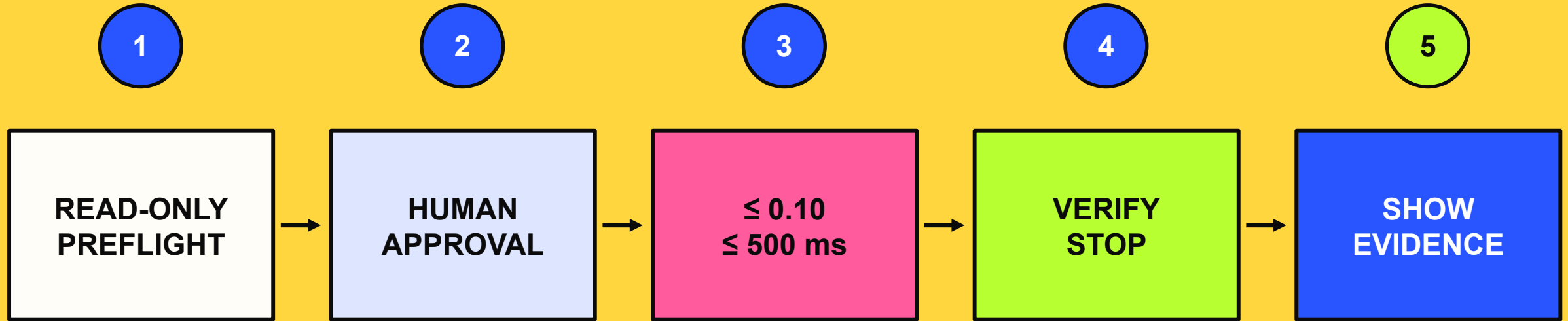
●● synthesizing... · 8s | deepseek v4 flash | 57.1k/1m
sdfgasd

LIVE PTY qualia_pinkie · 5 scoped tools
```

```
{
  "linear": 0.08,
  "angular": 0.0,
  "duration_ms": 400,
  "reason": "bounded demo"
}
```



Observe. Approve. Pulse. Prove zero.



ANY RED GATE → RECORDED FALLBACK

No improvising around the boundary.

NO AUTONOMY CLAIM

THE PLANNER PROPOSES.

THE BOUNDARY DECIDES.

Questions?

github.com/specdog/leash

<https://github.com/specdog/leash>



DECK + NOTES + EVIDENCE

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